Module 8 Mini-Project: Statistical Analysis using Orange

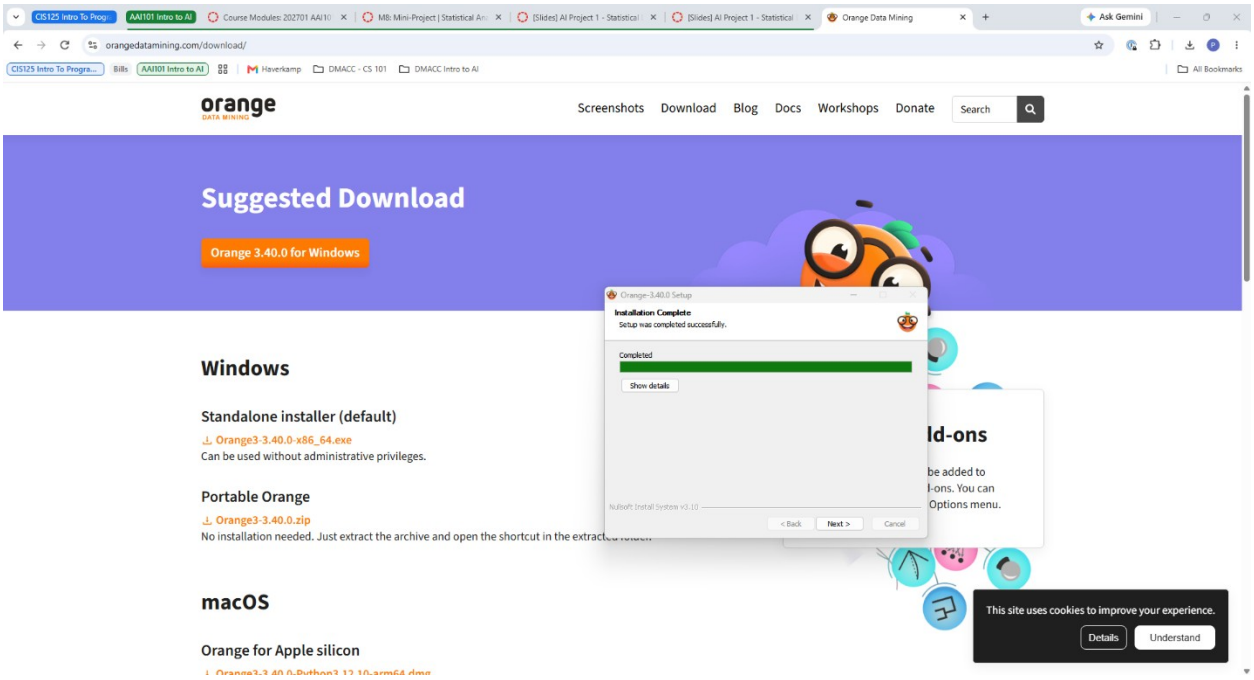
Patrick Hall
AAI101 Intro to AI
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AI was used for proofreading and formatting.

1. Installation & Setup

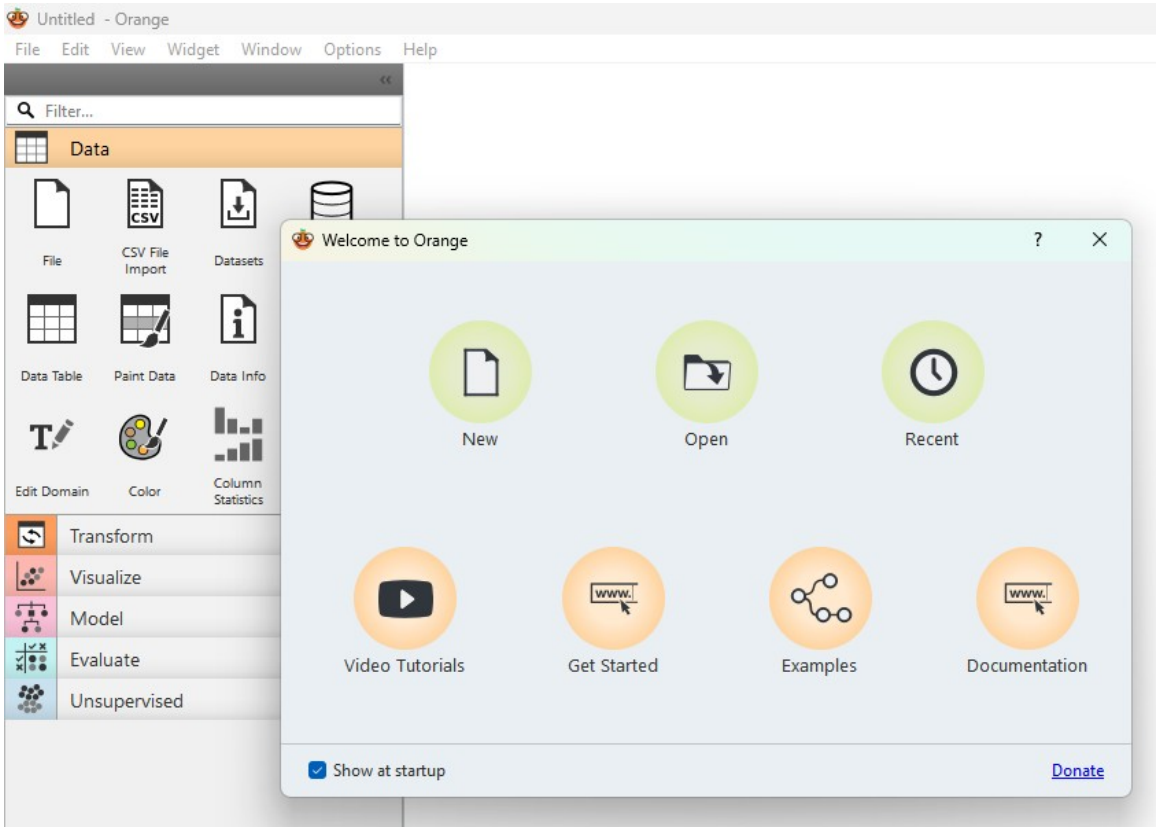
Downloading Orange

I downloaded the standalone Windows installer for Orange. While this was installing, I also went to Orange's YouTube video tutorials to help familiarize myself with the program's functions and features.



Opening Orange for the first time

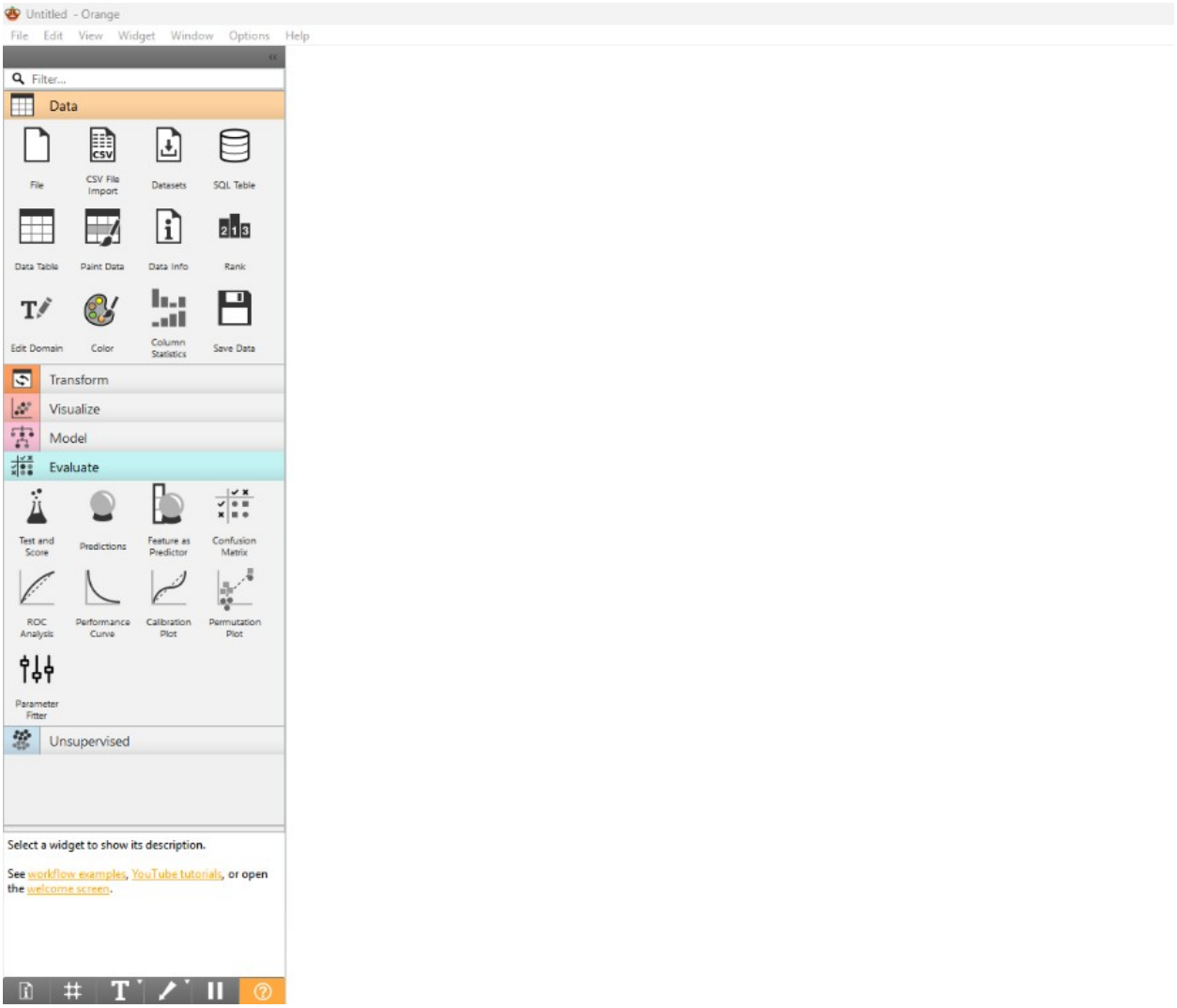
When I opened Orange for the first time, the Welcome to Orange screen came up with options like New, Open, Recent, and Video Tutorials. This showed me the install worked, and I was running Orange. I clicked New to start a blank workflow.



2. Orange Interface Familiarization

The workspace, widget panel, and canvas

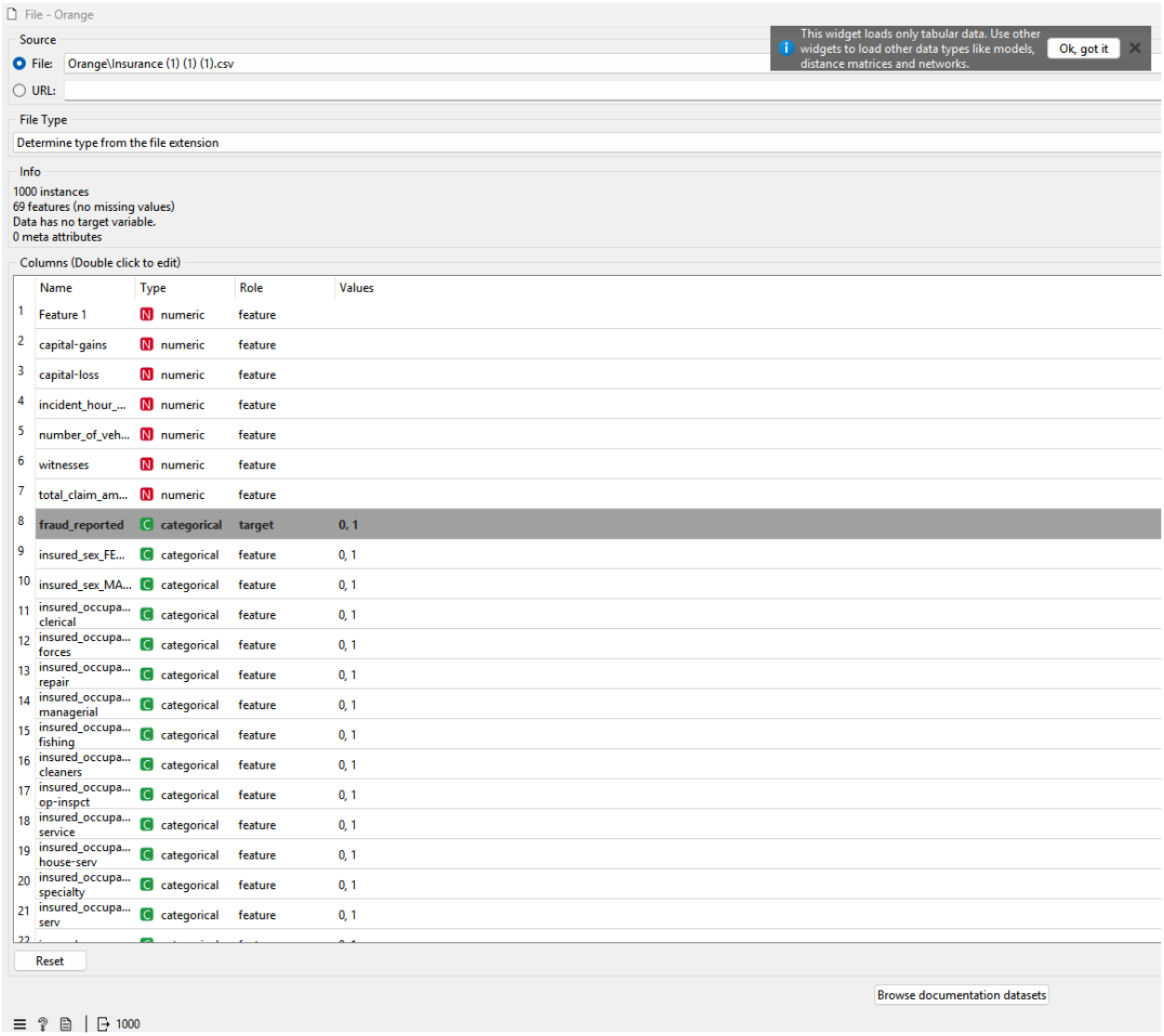
After clicking New, I got an empty workspace. The widget panel is the column on the left. It holds all the tools, grouped into categories like Data, Transform, Visualize, Model, Evaluate, and Unsupervised. The canvas is the big white area on the right, where you build a workflow by dragging widgets onto it.



3. Workflow Creation

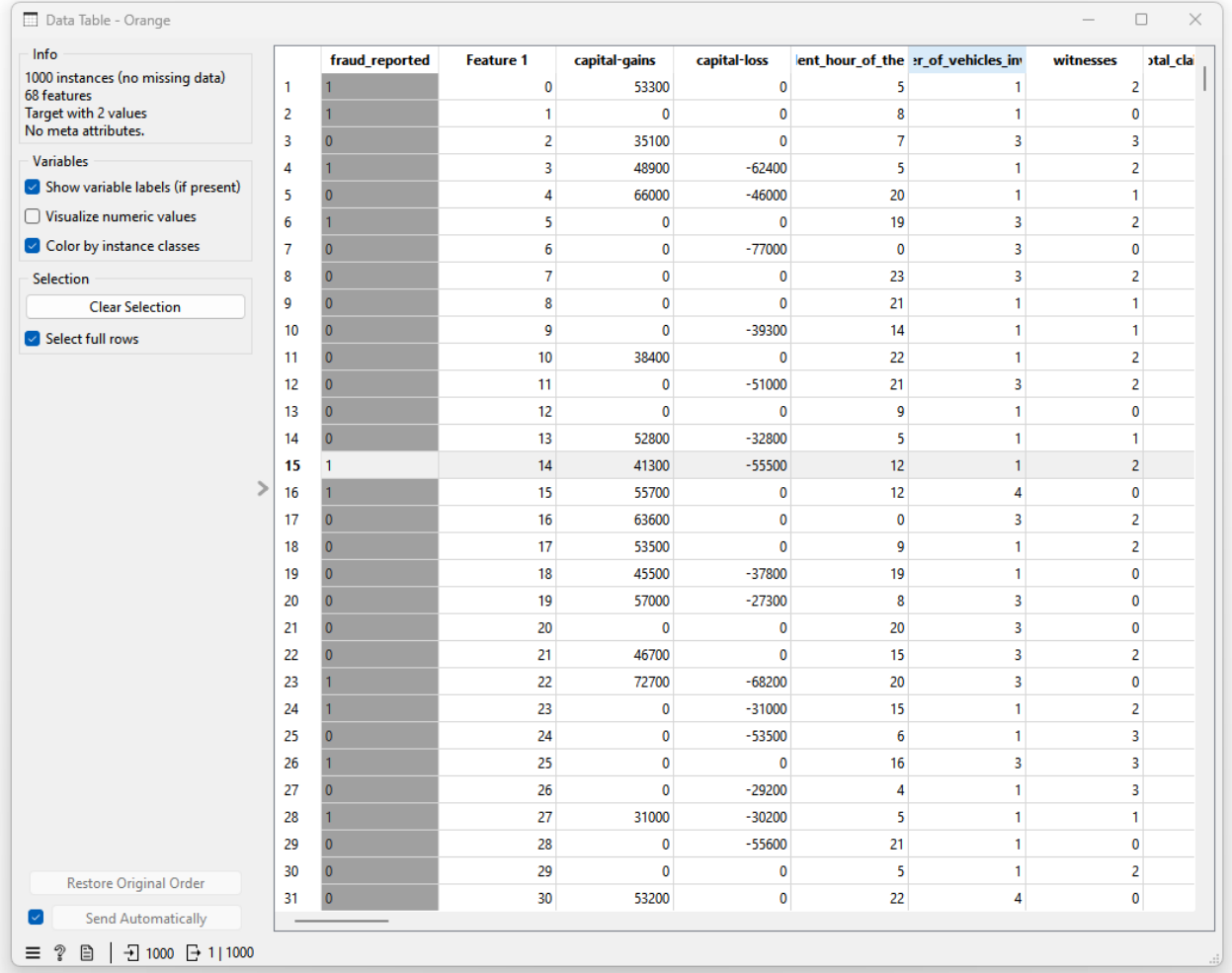
Loading Insurance.csv with the File widget

I dragged a File widget onto the canvas and used it to open Insurance.csv. Orange read 1000 instances and 69 columns with no missing values. I set fraud_reported as the target so Orange would treat it as the thing I wanted to study. It is a categorical column with the values 0 (no fraud) and 1 (fraud reported).



Checking the data in a Data Table

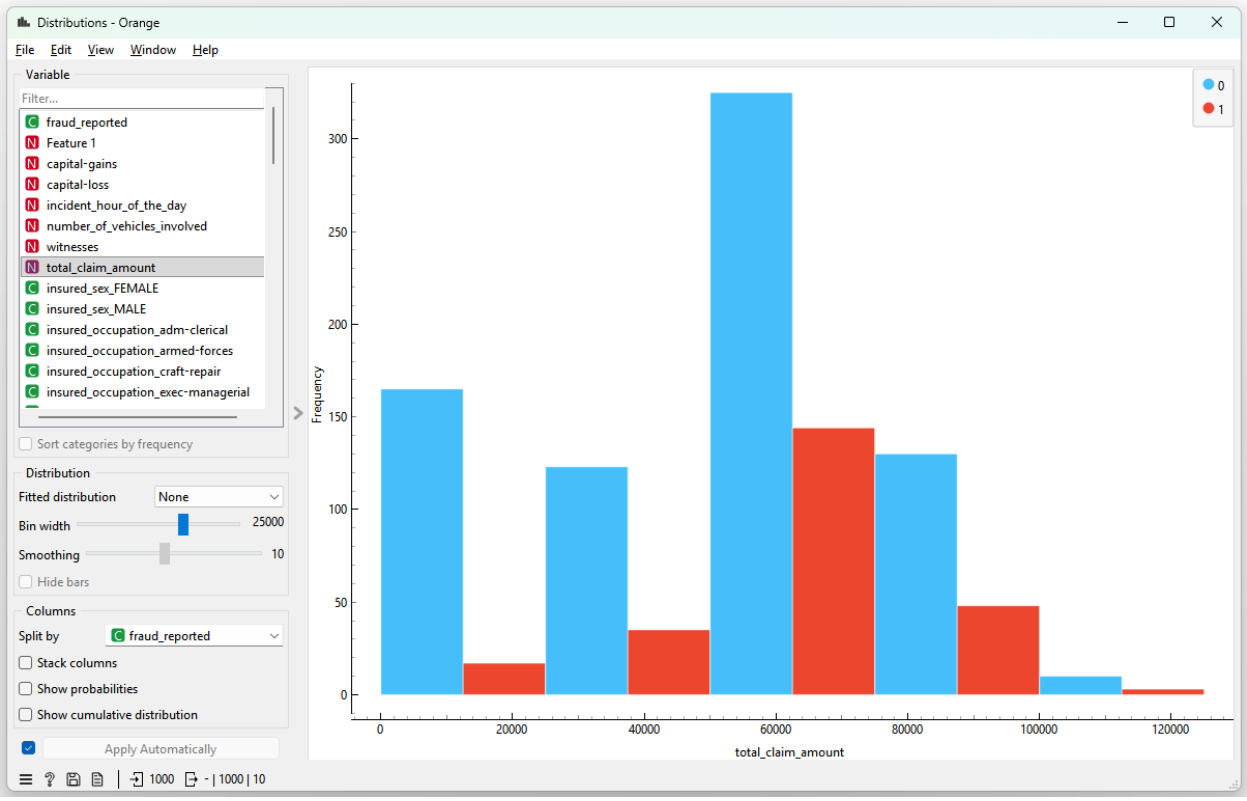
I connected a Data Table widget to the File widget so I could see the actual rows. It showed all 1000 rows, and the info panel confirmed the target has 2 values: 0 means not fraud and 1 means fraud. Out of the 1000 claims, 247 (24.7%) were fraud and 753 were not.



	fraud_reported	Feature 1	capital-gains	capital-loss	ient_hour_of_the	er_of_vehicles_in	witnesses	otal_cla
1	1	0	53300	0	5	1	2	
2	1	1	0	0	8	1	0	
3	0	2	35100	0	7	3	3	
4	1	3	48900	-62400	5	1	2	
5	0	4	66000	-46000	20	1	1	
6	1	5	0	0	19	3	2	
7	0	6	0	-77000	0	3	0	
8	0	7	0	0	23	3	2	
9	0	8	0	0	21	1	1	
10	0	9	0	-39300	14	1	1	
11	0	10	38400	0	22	1	2	
12	0	11	0	-51000	21	3	2	
13	0	12	0	0	9	1	0	
14	0	13	52800	-32800	5	1	1	
15	1	14	41300	-55500	12	1	2	
16	1	15	55700	0	12	4	0	
17	0	16	63600	0	0	3	2	
18	0	17	53500	0	9	1	2	
19	0	18	45500	-37800	19	1	0	
20	0	19	57000	-27300	8	3	0	
21	0	20	0	0	20	3	0	
22	0	21	46700	0	15	3	2	
23	1	22	72700	-68200	20	3	0	
24	1	23	0	-31000	15	1	2	
25	0	24	0	-53500	6	1	3	
26	1	25	0	0	16	3	3	
27	0	26	0	-29200	4	1	3	
28	1	27	31000	-30200	5	1	1	
29	0	28	0	-55600	21	1	0	
30	0	29	0	0	5	1	2	
31	0	30	53200	0	22	4	0	

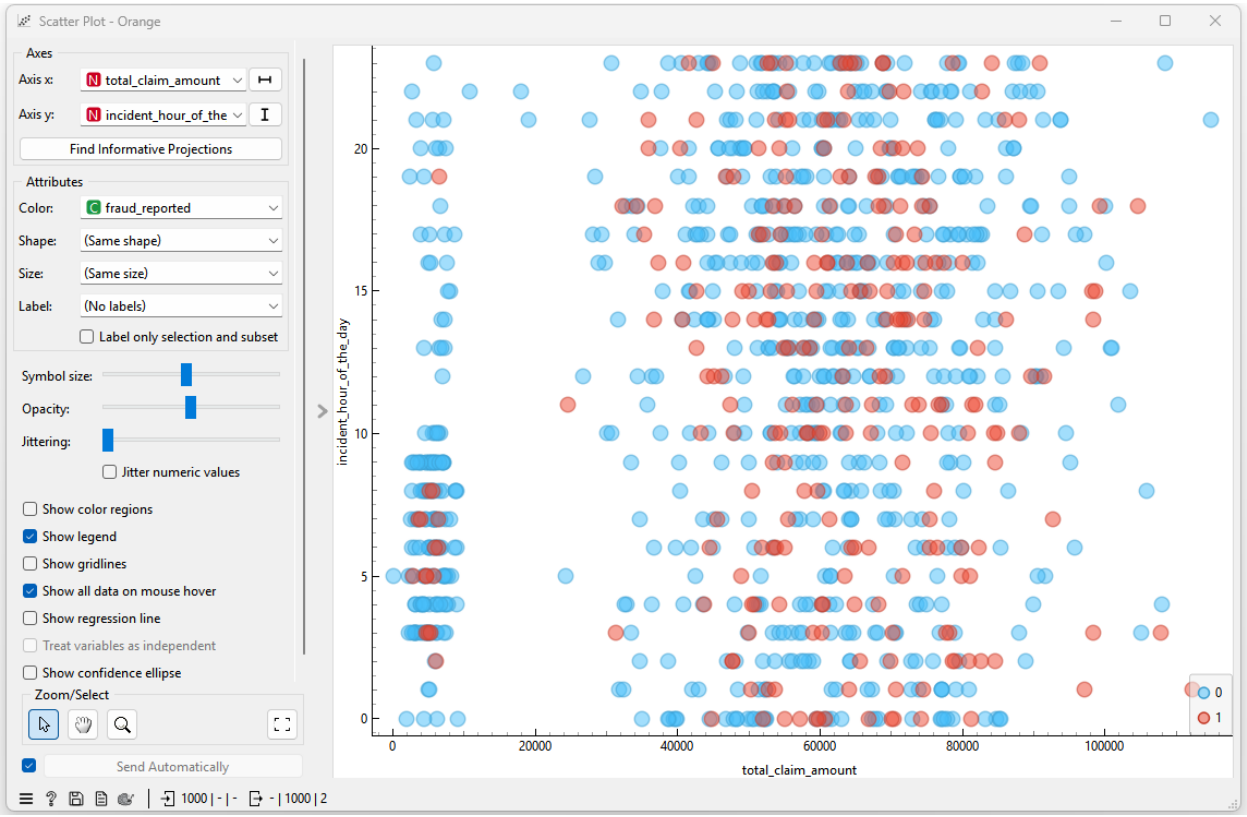
Distributions of total claim amount

Next, I added a Distributions widget, picked total_claim_amount, and split it by fraud_reported. Blue bars are non-fraud claims (0) and red bars are fraud claims (1). Most claims fall between about $25,000 and $75,000. The fraud claims lean toward the higher amounts and claims under about $25,000 are mostly non-fraud.



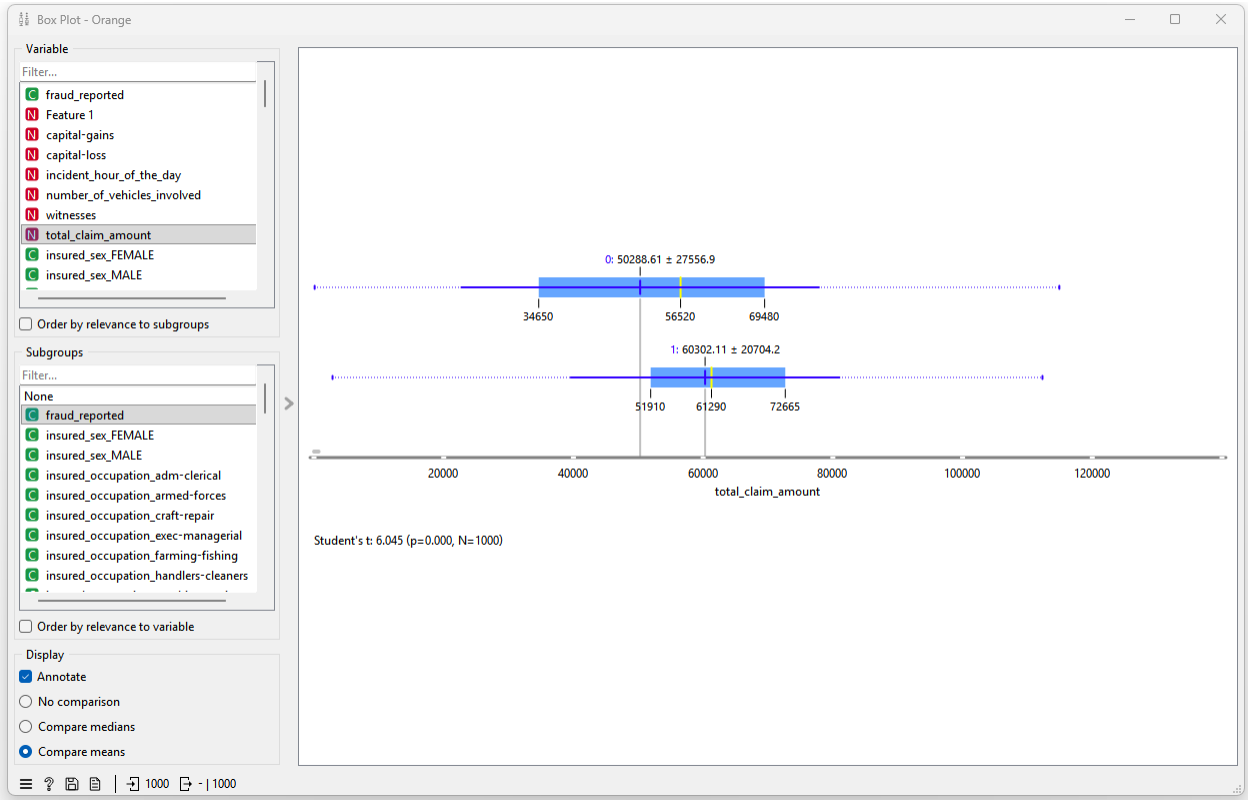
Scatter plot of claim amount vs. time of day

I added a Scatter Plot with total_claim_amount on the x-axis and incident_hour_of_the_day on the y-axis, colored by fraud. The red fraud dots show up at every hour of the day, so I did not see a time-of-day pattern. One thing that stood out is that the very smallest claims, under about $10,000, are almost all blue (non-fraud).



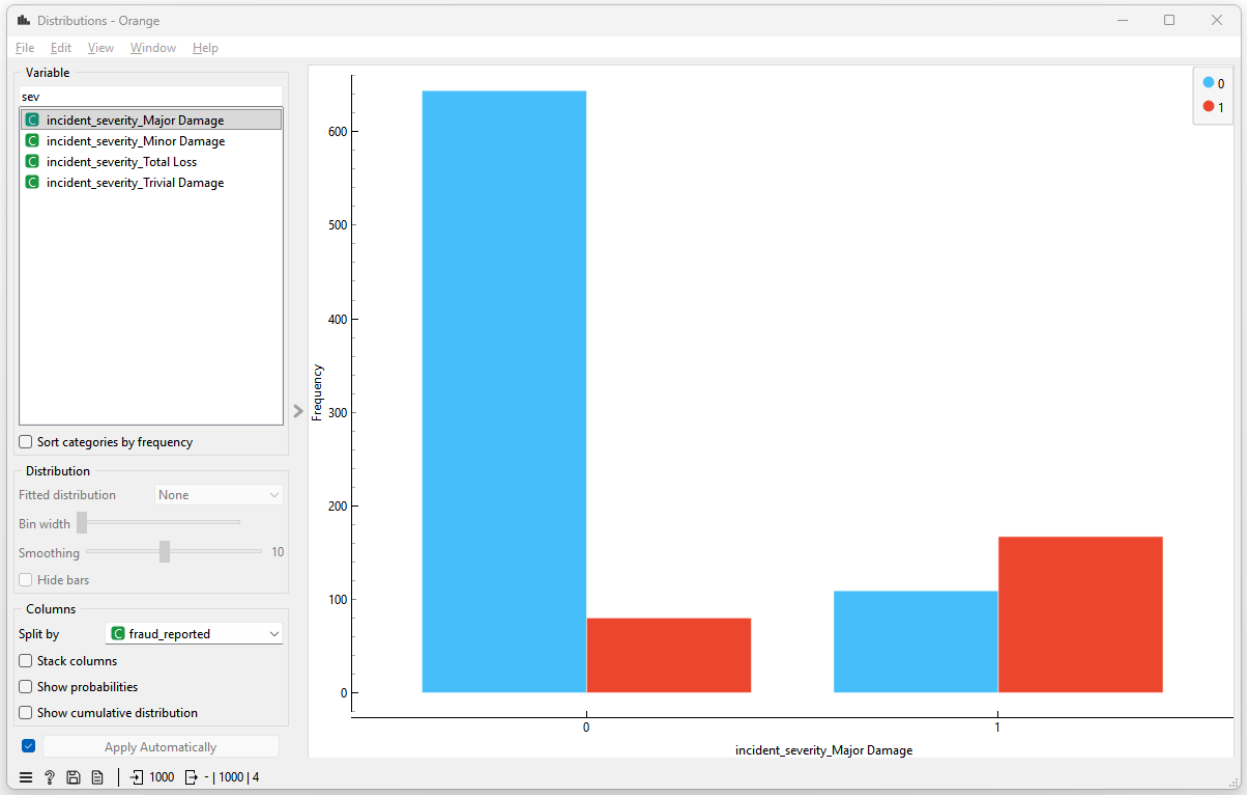
Box plot comparing claim amounts

Then I used a Box Plot of total_claim_amount grouped by fraud_reported, with Compare means turned on. The average non-fraud claim was $50,288.61 (SD 27,556.9) and the average fraud claim was $60,302.11 (SD 20,704.2), with medians of about $56,520 and $61,290. Orange's Student's t-test gave $p < 0.001$, so the difference is statistically significant.



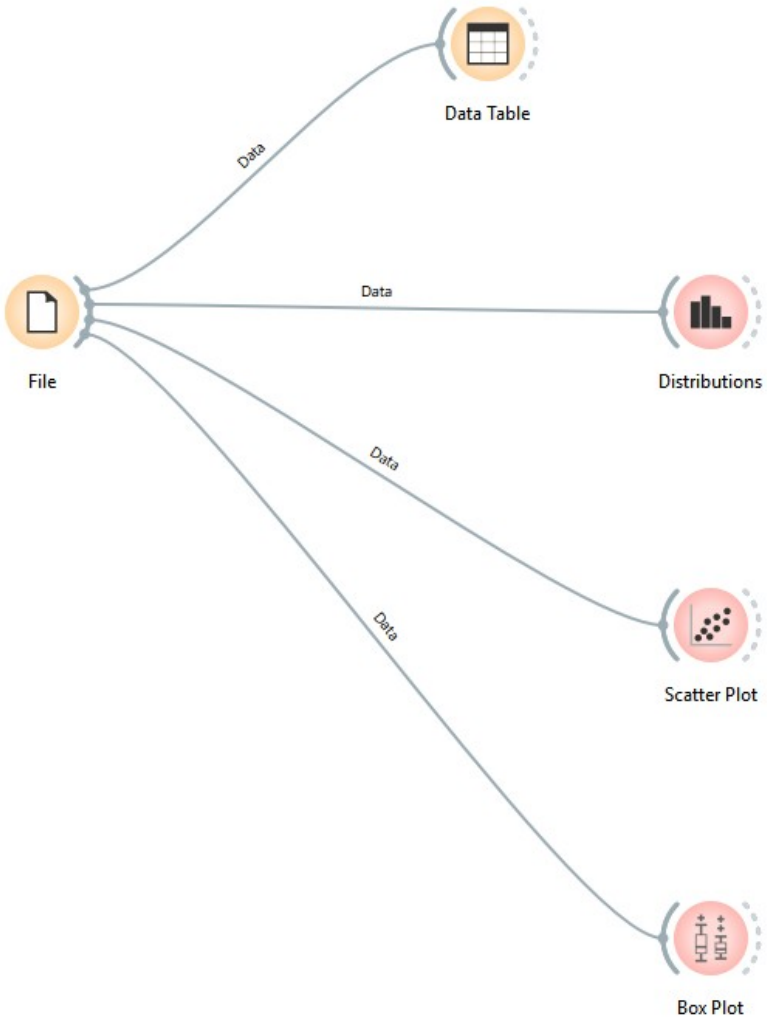
Fraud by incident severity

Next, I went back to the Distributions widget and picked incident_severity_Major Damage, split by fraud_reported, with Show probabilities ticked. The bar on the right (1) is the 276 claims with major damage, and 60.5% of them were fraud. The bar on the left (0) is every other claim, and only about 11% of those were fraud.



The finished workflow

This is my full workflow. The File widget feeds into four widgets: Data Table, Distributions, Scatter Plot, and Box Plot. Each gray line is a Data link going from the File widget's output to the input of the next widget. Having everything connected to one File widget made it easy to switch between different views of the same data. I used the same Distributions widget for both the claim amount chart and the severity chart by changing the variables.



4. Data Analysis & Interpretation

The first thing I found is that fraud claims tend to be bigger. On average, a fraud claim was about $10,000 more than a non-fraud claim ($60,302 vs. $50,288), and the t-test showed that difference is very unlikely to be just chance. Small claims were rarely fraud, especially the ones under about $10,000. On the other hand,

the hour of the day the incident happened did not seem to matter, because fraud showed up at all hours.

The strongest thing I found was incident severity. Here is how fraud broke down for each severity level in the CSV:

- Major Damage: 276 claims, 167 fraud (60.5%)
- Minor Damage: 354 claims, 38 fraud (10.7%)
- Total Loss: 280 claims, 36 fraud (12.9%)
- Trivial Damage: 90 claims, 6 fraud (6.7%)

Severity turned out to be a much stronger signal than claim amount. Fraud claims averaged about $10,000 more, but the fraud rate for major damage claims was more than double the 24.7% overall rate.

Here is a quick summary of what each view showed me:

- Data Table: 247 of the 1000 claims (24.7%) were fraud, so most claims in this dataset were not fraud.
- Distributions by Total Claim Amount: fraud claims lean toward higher claim amounts and claims under about $25,000 are mostly non-fraud.
- Scatter Plot: fraud shows up at every hour of the day, so the time of the incident did not point to fraud. This was not a good way of visualizing the data to detect fraud.
- Box Plot: the average fraud claim ($60,302.11) was higher than the average non-fraud claim ($50,288.61).
- Distributions by Severity: 167 of the 276 major damage claims (60.5%) were fraud, compared to 6.7% to 12.9% for the other severity levels.

For an insurance company, a practical takeaway would be to flag major damage claims for a closer review, since that is where most of the fraud shows up, and to give larger claims a second look too. Still, neither one can prove a claim is fraud, because 109 of the major damage claims and plenty of large claims were not fraud.

What I Learned

Using Orange showed me that I can do real data analysis without writing code, just by connecting widgets after loading data. Someone who isn’t a programmer can still play around with data and find patterns, which could help a business make better decisions.